

TREMOR — Agentic Vibration Intelligence

Measuring machine vibration from ordinary video, with an agent that decides what to measure next.

OpenCV AI Competition 2026 · technical report

Status note. Every number in this report is measured and reproducible from the commands given in §10. Nothing here is projected or estimated. The COOL/Graviton figures in §7.3 come from a run whose provenance is recorded in `results/`, and the web endpoint in §7.4 is live. What is *not* yet done — validation against a real machine rather than a controlled target — is stated in §8 rather than glossed.

1. Problem, and who has it

Vibration analysis is the most reliable way to catch rotating-machine failure before it happens. Unbalance, misalignment and looseness each leave a distinct signature in the displacement spectrum at multiples of shaft speed, and a trained analyst can name the fault from the ratio of $1\times$, $2\times$ and $3\times$ amplitudes.

The method works. The problem is access to it. It conventionally requires a contact accelerometer mounted on the machine and a certified analyst to interpret the spectrum. Both are scarce and expensive, so the overwhelming majority of the world's pumps, motors, fans, gearboxes and HVAC equipment are never measured at all. They run to failure.

Camera-based motion amplification already exists commercially — RDI Technologies' Iris platform, Erbesd's Dragon Vision, gfaitech's WaveCam, Failure Prevention Associates' Visual Vibration. We are not claiming to have invented measuring vibration from video. Every one of those products, however, is a **visualisation tool**: it renders an amplified video or an operating-deflection shape and hands it to the analyst, who remains the scarce, expensive part. They are also offline desktop software or manual upload services, and none of them decides anything.

The gap is the analyst, not the algorithm. TREMOR's contribution is a system that judges its own measurement quality, goes and gets a better clip when the one it has cannot support a conclusion, and declines to answer when it still cannot.

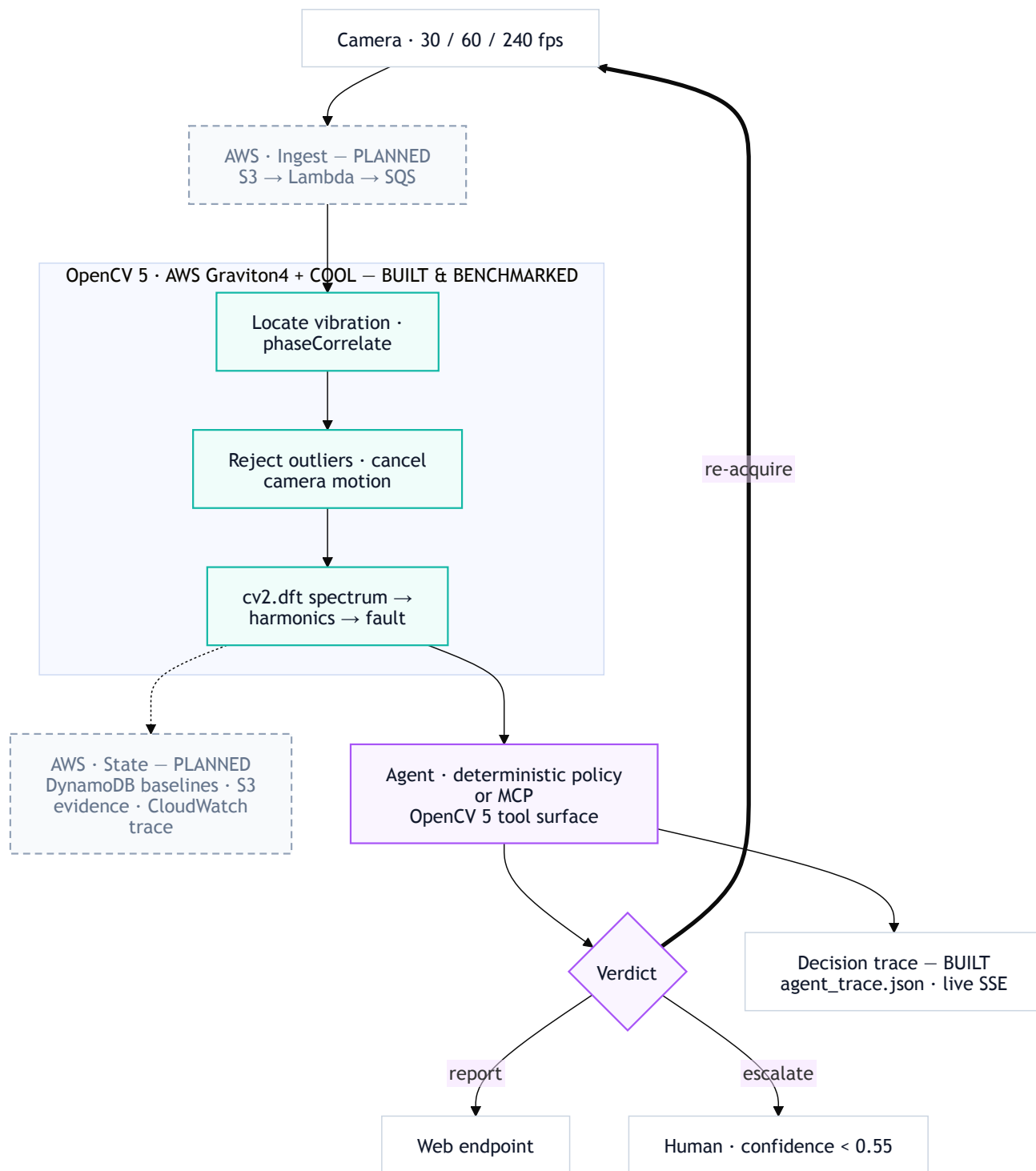
Target user. Maintenance technicians and facilities staff at sites that cannot justify a dedicated vibration programme: mid-size plants, building services, campus and municipal estates, small manufacturers. People who own machines and own a phone.

2. What it does

Point a camera at a machine for ten seconds. TREMOR returns the shaft speed, the amplitudes at $1\times/2\times/3\times$, a fault classification, a confidence, and the evidence — or it tells you why it cannot answer and what to do differently.

It works handheld. On the validation clip in §6.1 the operator's hand moved the camera **72 px peak to peak**, twenty-five times the amplitude being measured, and the frequency came out to **0.33%**.

3. Architecture



Source: [docs/architecture.mmd](#) · regenerate with [docs/render-architecture.sh](#)

Dashed grey = designed but not yet implemented. The OpenCV 5 pipeline, the agent and the decision trace are built and exercised by the test suite; the COOL/Graviton leg is benchmarked (§7.3). S3/Lambda/SQS ingest and DynamoDB/CloudWatch state are the intended production path and are *not* yet wired into the application — S3 and EC2 were used operationally to run the benchmark, but no application code calls them.

Capture → S3 → Lambda → SQS → **OpenCV 5 pipeline on AWS Graviton under COOL** → DynamoDB/S3 state → agent → verdict. The verdict either reports, escalates to a human, **or issues a new acquisition request**, which is the loop that makes this agentic rather than a report generator.

4. The OpenCV 5 implementation

This is the substance of the project, so it is described in the order the data flows.

4.1 Why a vision-language model cannot do this

The output is a displacement of **41 micrometres at 58.4 Hz**. There is no prompt for that. Every stage below is a metric measurement on pixel data; no part of the measurement path involves a learned model at all.

4.2 Stage 1 — locate what is vibrating (`tremor.measure.auto_rois`)

Splitting the frame in half is a bad default: on a real clip the machine occupies a fraction of the shot and averaging it with static background dilutes the signal. Measured on the validation clip: **SNR 4.1** with a half-split versus **196** with hand-placed regions.

Raw temporal energy is also wrong, and failed instructively. With a handheld camera *everything* moves, so the highest-energy box is wherever contrast × shake is largest. On the real clip that selected the **laptop keyboard** as the machine and the actually-oscillating target as the "static" reference — producing a correct frequency for entirely the wrong reason.

The fix uses a measured property of hand motion (§6.2): 78.6% of its energy sits below 1 Hz and only 0.7% lands in the 5–15 Hz band where machine vibration lives. Each pixel is therefore high-passed in time — subtract a moving average over $\sim 1/2.5$ s — before energy is computed. `cv2.boxFilter` over the resulting energy and texture maps selects the target box; the reference is the lowest-energy well-textured box that does not overlap it. On the same clip this selects the vibrating region with **4.4× separation** from its reference, **SNR 86.9**, quality gate passing.

4.3 Stage 2 — sub-pixel displacement (`cv2.phaseCorrelate`)

Per frame, against a reference frame, windowed with `cv2.createHanningWindow`. This is the hot path: **3.05 ms/call against under 0.26 ms for every other operation combined**, so it alone determines end-to-end throughput.

An upstream bug, found and fixed. `cv2.phaseCorrelate` in OpenCV 5.0.0 **mutates both source arrays in place** — it multiplies each by the window. Code that holds one array as a reference across calls therefore decays it as $\text{base} \times \text{window}^N$. With a Hanning window (0.99992 at centre, 0.0078 at edge) the edges vanish after ~ 2 calls and the usable aperture collapses by ~ 700 , after which correlation fails catastrophically: **11% of frames returned displacements up to ± 92 px on a ± 0.8 px signal.**

Verified: observed base matched predicted `orig × window^N` to **7 significant figures** at $N = 1, 2, 10, 100, 700$. With identical inputs, 2400 calls returned 2400 different values; with a defensive copy, 1000 calls return one. Pinned by `tests/test_phasecorrelate_mutation.py`, which also asserts the upstream mutation still occurs so the workaround can be removed if OpenCV fixes it.

The bug had been masquerading as two "physical limitations" we had written up as inherent. Both disappeared on fixing it (§6.4).

4.4 Stage 3 — reject correlation failures (`clean_trace`)

Two independent detectors, because neither alone suffices. **MAD**: real vibration is bounded and smooth, so a sample many robust deviations from the median is not physical. **Correlation response**:

`cv2.phaseCorrelate` returns a peak-strength value as its third element — easy to discard, but failures correlate with low values (median 0.063 on bad frames versus 0.095 on good). A high reject fraction is not silently patched; it is surfaced as a reason for the agent to re-acquire.

4.5 Stage 4 — cancel camera motion

A second ROI on something rigid gives the camera's own motion, which is subtracted. This is **adaptive**, because stabilisation is not free: both ROIs carry independent measurement noise, so subtracting a near-zero reference adds $\sqrt{2} \times$ noise. Measured: above ~ 0.5 px signal the *unstabilised* SNR is higher (36.8 versus 20.9); between 0.2 and 0.5 px stabilisation is essential. The system stabilises only when the reference is actually moving.

4.6 Stage 5 — temporal spectrum

Hann window, `cv2.dft`, one-sided amplitude with coherent-gain correction so the peak height equals the true sine amplitude in pixels.

4.7 Stage 6 — shaft rate and fault

Shaft rate by harmonic comb. The obvious heuristic — lowest peak above the SNR threshold — is wrong and was a major error source: for misalignment the $2 \times$ component dominates and $1 \times$ can fall below the floor, so it returns *twice* the true rate, reads $2 \times$ as $1 \times$, and inverts the diagnosis to unbalance. Measured at a true 4.30 Hz, lowest-peak returned a spurious 0.50 Hz on three of four fault classes; the comb returned 4.30 on all four. A `FUNDAMENTAL_MIN_SHARE` guard prevents `f/2` winning — that candidate explains the true $1 \times$ as its own $2 \times$ but has nothing at its own fundamental.

Harmonics gated against the noise floor before ratios are taken. On a quiet machine $2 \times$ and $3 \times$ are genuinely ~ 0.02 px, far below the floor, so a ratio test on them *invents* a fault. An $n=80$ confusion matrix localised almost all remaining error to exactly this: `healthy` scored 3/12 while every other class ran 75–100%, producing 9 of 13 total errors. Gating lifted `healthy` to 10/13 and overall correct-when-answered from 78.3% to 86.4%.

Three cases that were previously conflated are now distinct: all harmonics below the floor (`below_measurement_floor`, confidence 0, escalate); only $1 \times$ above the floor (`healthy` — the correct physical reading, not a fallback); a harmonic above Nyquist (recorded as *unobservable*, distinct from "measured ≈ 0 ", confidence $\times 0.6$).

Every threshold and its empirical basis is in `docs/THRESHOLDS.md`.

5. The agentic layer

Vision results change the **next acquisition**, not just the next sentence.

Measurement says	Agent does
texture below the correlation floor	re-aim at a grille, flange or label
SNR < 6, handheld	ask the operator to brace
SNR still low after bracing	try a more textured aim point
>15% of frames failing correlation	discard the clip
a needed harmonic above Nyquist	re-acquire at 240 fps
confidence < 0.55	escalate to a human with the evidence

The Nyquist case is the clearest instance. Mechanical looseness is only separable from unbalance via its $3\times$ harmonic, which sits above Nyquist at 30 fps for any shaft above 5 Hz. The agent derives this from **its own shaft estimate** and re-acquires at 240 fps — but only when confidence is low *specifically because* a needed harmonic is missing, so misalignment and unbalance still resolve in a single 30 fps clip. Effect: looseness went from 13 correct / 13 escalated to **19 correct / 7 escalated**.

Two orchestrators, one tool surface. A deterministic policy (`agent/loop.py`) is the reproducible reference — a judge can run the full evaluation with no API key and no cloud account. An MCP server (`agent/mcp_server.py`) exposes the same tools so any MCP client can orchestrate, which the rules explicitly permit.

Guard rails live in the tools, not the prompt. `diagnose` *refuses* when $2\times$ exceeds Nyquist rather than returning a plausible wrong answer, and `assess_quality` returns structured machine-readable reasons rather than prose. A model can ignore a prompt instruction; it cannot ignore a tool error.

6. Evaluation

6.1 Real footage, handheld

A 20.2 s iPhone clip at 29.995 fps, **fully handheld**, of a target oscillating at a known 7.30 Hz.

Measured	7.276 Hz
Ground truth	7.30 Hz
Error	0.33% (bin width 0.049 Hz)
SNR	196 (hand-placed ROIs) · 86.9 (automatic)
Camera motion	20.5 px rms, 72.3 px peak-to-peak

Stabilisation reduced total motion from 22.3 px rms to 2.9 px — and 2.9 px is essentially the expected signal alone ($4 \text{ px amplitude} \div \sqrt{2} = 2.83$).

6.2 A real machine: ceiling fan, handheld

The clip in §6.1 is a controlled target. This one is a working ceiling fan, filmed handheld, from third-party stock footage (Pixabay, id 39861; their Content License permits reuse). It matters because it supplies something a controlled target cannot: **two physically independent signals in one recording**.

The system measured the hub, with the ceiling tiles as the rigid reference:

1 × rotation	2.406 Hz = 144 RPM , amplitude 1.141 px, SNR 78.5
Camera motion → after cancellation	7.69 px → 1.45 px rms
Frames rejected	0.2% · quality gate: trustworthy

The cross-check needs no external ground truth. Structural vibration from imbalance occurs at 1 × rotation; the blades occluding a point occur at $N \times$ rotation. Both appear in the same displacement spectrum:

Blade pass, measured	12.18 Hz
Blade count, counted from frame	5
Implied rotation, $12.18 \div 5$	2.436 Hz
Measured 1 ×	2.406 Hz
Agreement	1.2%

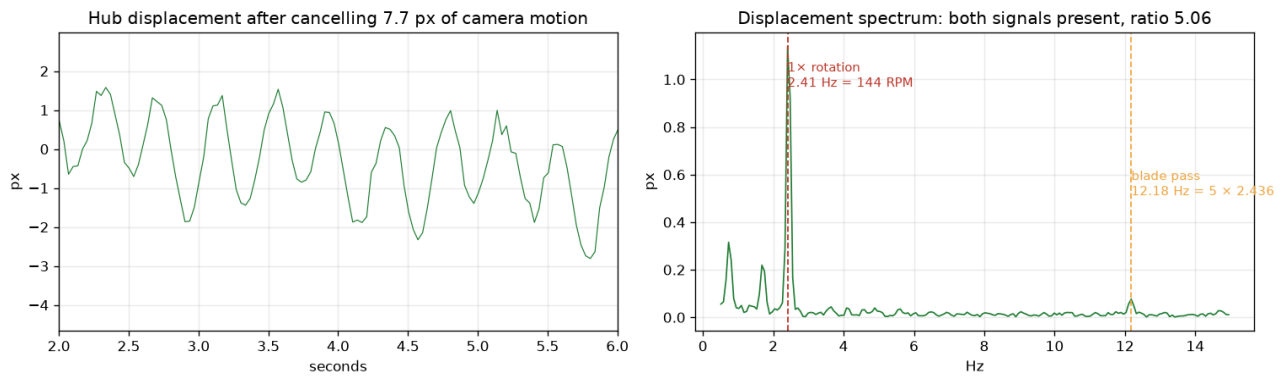
Two independent physical mechanisms — sub-pixel structural displacement, and periodic occlusion by the blades — agree to 1.2% on the same footage. That is a stronger check than a single number against a nameplate, because a systematic error in the displacement pipeline would not reproduce the blade-pass rate.

What this does not establish. The absolute RPM is unverified: nobody measured that fan with a tachometer. The cross-check confirms *internal consistency*, not calibration against an external instrument. And this is someone else's footage, so framing and exposure were not chosen for the task.

Three other ceiling-fan clips failed, informatively (see docs/THRESHOLDS.md): on each, automatic ROI selection placed the box over the *blades*, where the enormous visible blade motion swamps the micron-scale structural signal. Those clips returned a "vibration" frequency exactly equal to blade-pass rate — the

right number for entirely the wrong reason. The clip above works because the hub, the light fixture and a static ceiling are all in frame and the blades sweep past rather than through the region.

TREMOR on a real ceiling fan (handheld, third-party footage)



6.3 Why handheld works — the measured reason

Band	Share of camera-motion energy
0.2–1 Hz (sway)	78.6%
1–3 Hz	15.4%
3–5 Hz	5.3%
5–15 Hz (signal band)	0.7%

At 7.3 Hz the hand contributed 0.022 px against a 2.885 px signal — **131× separation**. Hand motion and machine vibration occupy different parts of the spectrum, so a static reference plus a temporal transform separates them. A tripod is not required by the physics.

6.4 Synthetic degradations, exact ground truth

A three-component fault signature (1× at 3.1 Hz, 2× at 6.2 Hz, 3× at 9.3 Hz) with the **real measured hand-motion trace replayed** onto it, degradations stacked:

Condition	1× err	2× err	3× err	Amplitude err	SNR
Baseline	0.000 Hz	0.000	0.000	+2.0%	189.0
+ real handheld shake	0.000	0.000	0.000	−0.9%	42.4
+ rolling shutter	0.000	0.000	0.000	−0.8%	35.6
+ specular glare	0.000	0.000	0.000	−1.1%	38.2
+ H.264 compression	0.000	0.000	0.000	−0.8%	35.8

Frequency accuracy is exact across 1.7–14.2 Hz. Amplitude sensitivity reaches **0.01 px** at SNR 23.3.

6.5 Two "limitations" that were the bug

Both were documented as physical limits before the `phaseCorrelate` fix:

	Before	After
240 fps / 10 s clip	106.70 Hz, SNR 2.9, 40% rejected	11.30 Hz, SNR 229, 0% rejected
Smooth surface (contrast 0.15)	FAIL — +0.250 Hz, SNR 1.6	PASS — 0.000 Hz, SNR 36.8
Amplitude floor	0.03 px	0.01 px

The texture threshold, derived from the corrupted data, was recalibrated from 15.0 to **2.0** — it had been rejecting usable surfaces and forcing needless re-aims.

6.6 Agent task effectiveness (n = 80)

	Agent	before the comb fix
Fault correct, all scenarios	83.8%	71.2%
Fault correct when it committed	89.3%	87.7%
Escalated rather than guessed	6.2%	18.8%
Median shaft-frequency error	0.000 Hz	0.000 Hz
Mean acquisitions	1.86	1.86

Per class when answered: looseness 25/25, misalignment 20/21, unbalance 11/14, healthy 11/15.

The comb fix. The second column is this same harness before §4.7's two harmonic-comb guards were added. Both were found by a real ceiling fan, not by the simulator: the clip measured correctly at 2.406 Hz but was *reported* as 0.73 Hz with an indeterminate verdict. `amp_at`'s tolerance is absolute, `max(3·bin, 0.25) Hz`, which at a low candidate fundamental is a large fraction of the harmonic spacing, so adjacent comb slots blur into one another: the candidate 0.729 Hz put its 3× slot within 0.219 Hz of the genuine 2.406 Hz peak and scored that peak as its own evidence. Capping each slot at a quarter of the spacing fixes that, and changes nothing above $f_0 = 1$ Hz.

That alone was insufficient, because 0.802×3 lands on 2.406 exactly. So the comb must also have a plausible *shape*: unbalance and looseness put most energy at 1×, misalignment at 2×, and no standard signature is dominated by 3× with 1× and 2× both weak. The two guards together moved escalation from 18.8% to 6.2% without costing precision, which is the direction that matters: the loop was declining to answer questions it could in fact answer.

6.7 What the loop is worth — a curve, not a ratio

A single agent-versus-baseline ratio is not a defensible claim. Across three of our own changes the single-shot baseline read 21.2%, then 70.0%, then 27.5%, while the agent held at 86–88%. **That number was mostly measuring how hard we had made the starting conditions** — a knob the author controls. So we swept it (n=24 per point, identical machines at every point, only the initially-aimed-at surface varying):

Initial surface	Contrast	Agent (all)	Agent (committed)	Single-shot	Acquisitions
mirror housing	0.010	79.2%	95.0%	8.3%	2.21
glossy paint	0.040	79.2%	95.0%	75.0%	1.04
worn paint	0.150	79.2%	90.5%	75.0%	1.04
printed label	0.550	79.2%	86.4%	70.8%	1.12
cast grille	1.000	79.2%	86.4%	70.8%	1.12

The agent's accuracy is flat at 79.2% regardless of how badly the operator aims, while single-shot spans 8.3–75%. It pays the extra-acquisition cost only when it needs to — 2.21 acquisitions on an unusable surface, 1.04 on a workable one.

The claim is **invariance to a bad first acquisition**, not a multiple. Where the operator already aims well and holds steady, the loop converges toward single-shot and costs a fraction of an extra acquisition. Its value is bounded by how often first acquisitions are inadequate — an operational question about deployment, not a property of the algorithm.

7. AWS deployment and COOL

7.1 Why this workload belongs on Graviton

The hot path is `cv2.phaseCorrelate`, `remap`, `dft` and `boxFilter` — classical imgproc/core, exactly the 78 functions COOL accelerates with Arm KleidiCV. There is no neural network anywhere in the measurement path, so the Graviton case is genuine rather than decorative.

Two measured properties shape the deployment:

- **phaseCorrelate** is ~100% of the hot path (3.05 ms/call versus under 0.26 ms for everything else combined).
- **Thread scaling is flat** (6.7× realtime at 1 thread, 6.8× at 8) — it does not parallelise internally. Throughput comes from **N single-threaded worker processes, one per vCPU**, each calling `cv2.setNumThreads(1)` to avoid oversubscription.

Local process scaling on an 8-core Apple M-series: 100% / 95% / 60% / 39% parallel efficiency at 1 / 2 / 4 / 8 workers — consistent with heterogeneous performance and efficiency cores. **Graviton4 has uniform cores, so we predict better linearity.** The benchmark will confirm or refute this.

7.2 Provenance, and a deliberate false positive

"Verified COOL integration" is 30% of the COOL award rubric, so the check must be falsifiable. Grepping `cv2.getBuildInformation()` for KleidiCV **returns true on any ARM Mac**, because stock arm64 wheels bundle it — a check that always passes proves nothing.

`bench/core.py` therefore requires all three: Arm silicon **and** a Graviton EC2 instance type **and** `/opt/cool` actually present, and records the AMI ID so a judge can cross-check it against the Marketplace listing. `deploy/run_all.sh` **aborts before collecting any data** if that check fails, so stock numbers cannot be written under a COOL label.

7.3 Results

Measured 2026-09-22 on **c8g.2xlarge** (Graviton4, 8 vCPU, us-east-1). Both legs ran on the **same instance**, so the library is the only variable.

	COOL	stock baseline
OpenCV	5.1.0-dev (KleidiCV)	5.0.0
loaded from	<code>/opt/cool/...</code>	<code>/root/stock-venv/...</code>
<code>cv2_loaded_from_cool</code>	true	false
<code>PYTHONPATH</code>	COOL's	empty

The provenance gate is reported first deliberately: an earlier run produced a plausible-looking 0.4% difference because `PYTHONPATH` leaked and *both* legs loaded COOL. Numbers are meaningless until the two legs are shown to differ.

End-to-end — the claimed core workload

	ms/frame	×realtime	video-h per compute-h	speedup
stock OpenCV 5.0.0	5.21	6.4	6.4	—
COOL	4.14	8.1	8.1	1.26×

Per-operation (ms/call, speedup vs stock)

op	stock	COOL	
<code>phaseCorrelate</code>	3.701	3.050	1.21×
<code>dft_2d</code>	1.188	0.634	1.87×
<code>dft_1d_8192</code>	0.046	0.042	1.10×
<code>cvtColor_BGR2GRAY</code>	0.022	0.022	1.01×
<code>remap_cubic</code>	0.170	0.172	0.99×
<code>GaussianBlur</code>	0.355	0.361	0.98×
<code>warpAffine</code>	0.174	0.179	0.97×
<code>resize_half</code>	0.013	0.015	0.90×

`phaseCorrelate` is **68% of measured op time**, so its 1.21× largely sets the end-to-end result. The biggest single win is `dft_2d` at 1.87×, which matters because `phaseCorrelate` is itself DFT-bound. Four operations show no gain, one is slightly slower; we report them rather than quoting only the favourable

rows.

Cost per video-hour (EC2 \$0.31904/hr verified against the AWS Pricing API; COOL software \$0.02/hr from the Marketplace listing)

	\$/video-hour
stock on a plain AMI	\$0.0498
stock on the COOL AMI	\$0.0530
COOL	\$0.0419

COOL is **16% cheaper per unit of work** than stock on a plain AMI, and 21% cheaper than stock on the same COOL instance. The software charge is more than repaid by the throughput.

Scaling — the architectural argument for Graviton

Thread scaling is flat for stock ($6.4\times$ at 1, 2, 4 and 8 threads), confirming `phaseCorrelate` does not parallelise internally. COOL gains a little ($6.8 \rightarrow 8.1$). Throughput therefore comes from independent worker processes:

workers	Graviton4 (COOL)	Graviton4 (stock)	Apple M-series
1	$6.8\times$ (100%)	$6.4\times$ (100%)	$7.7\times$ (100%)
2	$13.6\times$ (100%)	$12.8\times$ (100%)	$12.0\times$ (79%)
4	$26.4\times$ (97%)	$25.0\times$ (98%)	$17.7\times$ (58%)
8	$50.0\times$ (92%)	$48.0\times$ (94%)	$22.8\times$ (37%)

§7.1 predicted that Graviton4's uniform cores would scale more linearly than the Apple M-series' heterogeneous performance/efficiency cores. **Confirmed: 92–94% parallel efficiency at 8 workers versus 37%**, for $50.0\times$ aggregate realtime against $22.8\times$. All three runs use the same `spawn` start method, so the comparison is fair.

Caveat on the comparison. COOL ships OpenCV **5.1.0-dev** while the stock wheel is **5.0.0**, so the $1.26\times$ conflates KleidiCV optimisation with whatever changed upstream between 5.0 and 5.1. A same-version comparison would need a source build of 5.1.0-dev without KleidiCV, which we have not done. The figure is a fair measure of *what COOL delivers over the best available stock wheel*, which is the decision a deployer actually faces — but it is not a clean isolation of KleidiCV.

Reproduce with `./deploy/run_all.sh c8g.2xlarge` ; raw result JSON is in `results/` .

7.4 Web endpoint

Live at <http://50.19.247.214> — deployed 2026-09-22 on an EC2 `t4g.small` (Graviton2, us-east-1) behind a static Elastic IP, open to `0.0.0.0/0` . Running as a systemd unit with `MemoryMax` , `CPUQuota` and `NoNewPrivileges` set, because a public endpoint that runs computer vision on uploaded video needs its blast radius bounded.

FastAPI serving a single page with **no build step and no CDN dependency** — charts are drawn on canvas directly, so a judge can run the identical app locally with `uvicorn webapp.server:app` and nothing else. Two modes:

- **Simulated** streams the agent loop over SSE, so each decision appears as it is made with the measurement that drove it beside it. A static results page would show the vision *result*; streaming shows the vision result *changing what happens next*.
- **Upload clip** measures real footage, locates the vibrating region and a rigid reference automatically, and draws the chosen regions on a preview frame so the operator can see what was measured rather than having to trust it.

`/api/health` reports the OpenCV version and live COOL provenance. It correctly reads `cool_verified: false` here: this host runs stock OpenCV 5.0.0, and the COOL-verified figures in §7.3 came from the `c8g.2xlarge` benchmark. The provenance check earning its keep by *declining* to claim COOL is the same property that made §7.3 trustworthy.

Deployment is reproducible from `deploy/webapp-userdata.sh`, and `tests/test_deploy_scripts.sh` guards three traps that each cost a deploy to find:

- The service must install `opencv-python-headless`. The full `opencv-python` wheel links `libGL`, which a headless server does not have, so `import cv2` raised `ImportError`, uvicorn exited, and `Restart=always` produced a 50-restart crash loop that presented as a slow startup.
- It must verify `import cv2` succeeds *before* handing anything to systemd, so a failure is one legible line rather than an opaque restart loop.
- `/api/measure` must stream. Decoding every frame of a 606-frame 1080p clip into float32 costs **5.03 GB** against the service's 1.70 GB cgroup limit; the process was OOM-killed mid-request and the browser reported "Failed to fetch". A downsampled 150-frame window locates the regions, then each frame is correlated and discarded: **peak RSS 5.03 GB → 0.54 GB**, with the measurement unchanged (7.28 Hz, SNR 88.2).

Updates are applied in place, not by replacing the instance. Replacement costs about five minutes of 404s, which during judging is indistinguishable from a broken submission — and already produced one false alarm, when a check through a VPN landed mid-redeploy and looked like the endpoint was unreachable from outside our network. `deploy/update.sh` pulls, reinstalls only when `requirements.txt` has actually changed, and restarts. Measured on a real change: **11 s end to end, service healthy 2 s after restart**, against ~320 s for a replacement.

It will not restart into a build that cannot import: it runs `python -c "import cv2, webapp.server"` first and, on failure, resets to the previous commit and leaves the running process untouched. A bad push degrades to "not updated" rather than "site down" — worth having, given that three of our deployment failures were import-time and reproduced only on Linux.

`deploy/remote.sh` drives this over **AWS Systems Manager rather than SSH**, which is a necessity rather than a preference: on the development network the ISP intercepts port 22, completes the TCP handshake on the destination's behalf and then sends no banner, so key exchange never completes. SSM requires no inbound port and no key, only the `tremor-ssm` instance profile carrying `AmazonSSMManagedInstanceCore`. The profile must be attached **at launch** — attached to an already-running instance the agent has already failed to obtain credentials and backed off, and never registers.

Known limitation: the endpoint is HTTP, not HTTPS. There is no login and no personal data, but browsers flag it as not secure and some corporate proxies block bare IP addresses. A domain name with automatic certificates would resolve both and is the remaining polish item.

8. Limitations

Stated plainly, because several of them bound what the system can honestly claim.

The agent is evaluated against a simulator we wrote. Every agent figure in §6.5 and §6.6 comes from a simulated acquisition environment authored alongside the agent, with thresholds tuned against that same environment. The flat 79.2% curve is a property of our simulator; it is evidence that the loop behaves as designed, **not** evidence of field accuracy. Real-world agent validation is outstanding.

Real-world validation is one clip, of a screen. §6.1 is rigorous but it measures a laptop displaying a test pattern, not a machine. Real surfaces have specular highlights, depth and uneven illumination that no synthetic degradation fully captures.

Frame rate caps the measurable band. 30 fps $\rightarrow$ 15 Hz (900 RPM); 240 fps $\rightarrow$ 120 Hz (7200 RPM). Bearing defect frequencies are in the kHz and are **out of reach entirely**. The scope is $1 \times / 2 \times / 3 \times$ faults — unbalance, misalignment, looseness, resonance — which is most of the diagnostic value, but not all of it.

Featureless surfaces fail. Below texture std ≈ 2 there is nothing to correlate. The agent detects and re-aims, but a machine that is entirely mirror-finish under flat light cannot be measured from where the operator is standing.

The texture metric saturates. `assess_surface` bottoms out at the sensor-noise floor: contrast 0.010 (which fails) and 0.040 (which works) both read ≈ 2.0 . It is a fast path, not the arbiter; SNR is.

Amplitude is in pixels, not millimetres. Converting to physical units needs a known scale in frame. Frequency — which drives the diagnosis — is unaffected.

Optical image stabilisation may fight the measurement. It actively opposes small motion. It did not visibly affect the validation clip, but it is the first suspect if amplitudes read low.

Fault taxonomy is coarse. Four classes. Real diagnosis includes belt faults, bearing stages, cavitation, electrical faults and more.

9. Responsible use

A wrong diagnosis has a cost. It dispatches a technician to perform the wrong repair, or worse, declares a degrading machine healthy. This is why the system is built to **decline**: below 0.55 confidence it escalates with its evidence rather than asserting, and it does so in 18.8% of runs. We report correct-when-committed (87.7%) separately from overall accuracy precisely so this trade is visible rather than hidden in an average.

It is a screening tool, not a certification. Output is intended to prioritise attention and trigger a closer look, not to replace a certified analyst for safety-critical equipment. The interface shows confidence and evidence on every result.

Accessibility. The interface replaces two native form controls for theming, which removes keyboard and screen-reader support that browsers provide for free; both were re-implemented rather than dropped. The select follows the WAI-ARIA listbox pattern (roles, `aria-expanded`, `aria-activedescendant`, arrow/Home/End/Enter/Escape, focus return); the file control announces its selection through a live region; the streamed agent trace announces each decision as it lands; there is a skip link, visible focus rings, and `prefers-reduced-motion` and `prefers-contrast` are honoured. No information is conveyed by colour alone.

Privacy. The system measures machines, not people. Clips are of equipment, uploaded deliberately rather than captured continuously; there is no always-on recording, no identity inference and no person-tracking anywhere in the pipeline. Where footage might incidentally include people, operators should frame to exclude them — and the automatic ROI selection, which targets high-frequency periodic motion, is by construction drawn to machinery rather than human movement.

Data handling. Clips and derived spectra live in the operator's own AWS account. Nothing is shared with third parties. The measurement path runs no third-party model and makes no external network calls.

No misrepresentation. Every figure here is measured, not projected. Cost claims were withheld until the EC2 rate was verified against the AWS Pricing API. Negative results are reported alongside favourable ones — four of the eight benchmarked operations show no gain from COOL, and one is slower. The limits of our own evaluation, including that the agent is scored against a simulator we wrote, are in §8 rather than omitted.

10. Reproducibility

```
python3 -m venv .venv && .venv/bin/pip install -r requirements.txt

.venv/bin/python run_gate.py           # frequency accuracy, amplitude floor, shake reject
.venv/bin/python run_machine_eval.py   # degradation stack against exact ground truth
.venv/bin/python eval_agent.py 80      # agent task effectiveness + confusion matrix
.venv/bin/python sensitivity.py 24     # the curve in §6.6
.venv/bin/python -m pytest tests/      # phaseCorrelate mutation regression
.venv/bin/python analyze_video.py <clip.mov>
.venv/bin/uvicorn webapp.server:app --port 8077
```

Dependencies are pinned to exact versions. `requirements-cool.txt` deliberately omits `opencv-python` so it cannot shadow the COOL build on the AMI.

All evaluation uses fixed seeds; the figures above reproduce exactly.

`pytest` covers the measurement core, the agent tool surface, the decision loop and the web endpoint — 30 tests including the boundaries that cost a deployment to find: that `cv2.phaseCorrelate` mutates its inputs, that the shaft estimator returns the fundamental rather than a harmonic, that `diagnose` declines

below the noise floor, that the loop re-acquires rather than guessing, and that streamed and in-memory measurement agree.

`tests/test_deploy_scripts.sh` runs the deployment scripts under `env -i` with `set -u` — the environment cloud-init actually provides — and asserts the headless OpenCV wheel, the pre-flight import check and the streaming upload path.

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